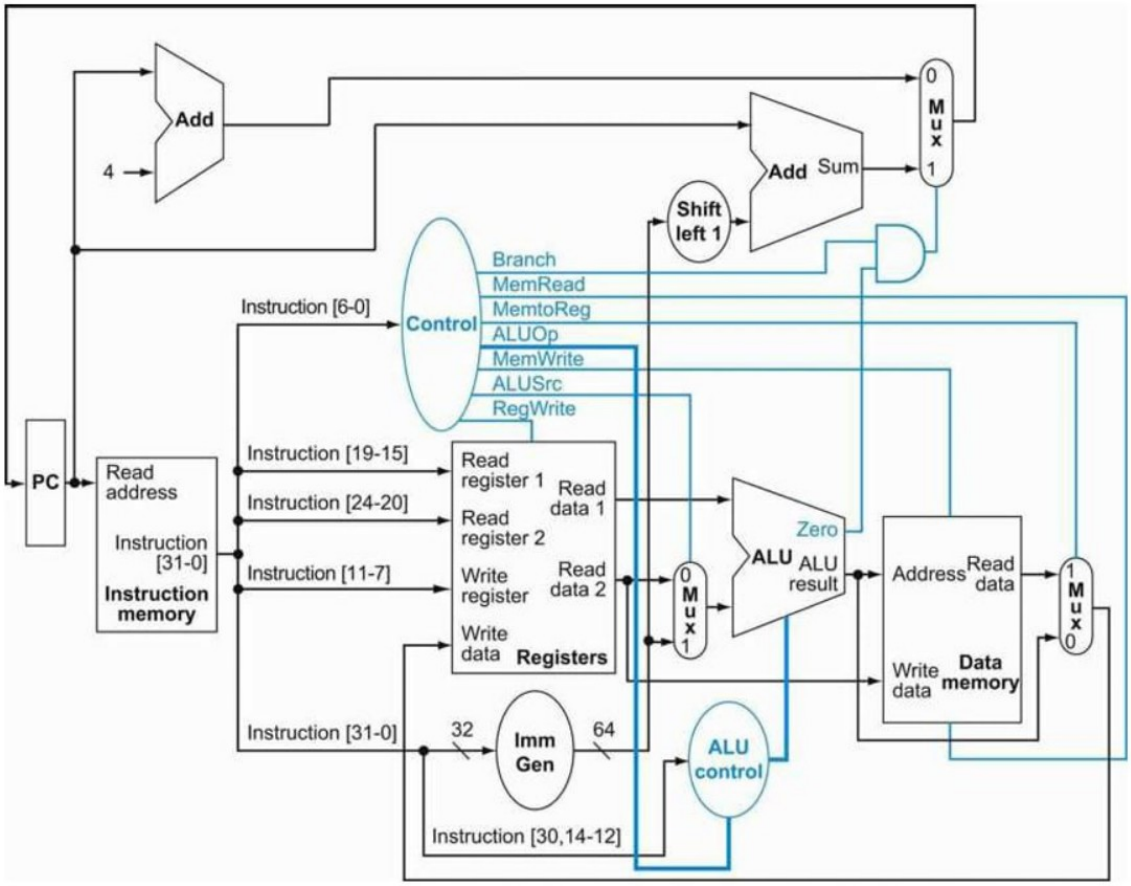
۱- dataPath

ساختار DataPath



۲- نتایج

۲-۱- پیدا کردن ماکسیمم

```c
#include <stdio.h>

int main() {
   int arr[20] = {1,4,7,2,8,11,3,15,6,9,12,14,10,13,5,0,2,3,4,8};

   int max_index, max;
   max_index = 0;
   max = arr[0];

   for (int i = 1; i < 20; i++) {
      if (arr[i] > max) {
         max = arr[i];
```

```
            max_index = i;
        }
    }

    printf("The maximum value is %d at index %d\n", max, max_index);

}
```

کد اسمبلی:

```
  addi sp, zero, 0
  addi t0, zero, 5
  sw t0, (0)sp
  addi t0, zero, 4
  sw t0, (4)sp
  addi t0, zero, 7
  sw t0, (8)sp
  addi t0, zero, 2
  sw t0, (12)sp
  addi t0, zero, 8
  sw t0, (16)sp
  addi t0, zero, 11
  sw t0, (20)sp
  addi t0, zero, 3
  sw t0, (24)sp
  addi t0, zero, 15
  sw t0, (28)sp
  addi t0, zero, 6
  sw t0, (32)sp
  addi t0, zero, 9
  sw t0, (36)sp
  addi t0, zero, 12
  sw t0, (40)sp
  addi t0, zero, 14
  sw t0, (44)sp
  addi t0, zero, 10
  sw t0, (48)sp
  addi t0, zero, 13
  sw t0, (52)sp
  addi t0, zero, 5
  sw t0, (56)sp
  addi t0, zero, 0
  sw t0, (60)sp
  addi t0, zero, 2
  sw t0, (64)sp
  addi t0, zero, 3
  sw t0, (68)sp
  addi t0, zero, 4
  sw t0, (72)sp
  addi t0, zero, 8
  sw t0, (76)sp


  add s0, zero, zero // max_inedx
  lw t0, (0)sp
  add s1, t0, zero // max
```

```
  addi s2, zero, 1 // i
  addi s3, zero, 4 // 4*i

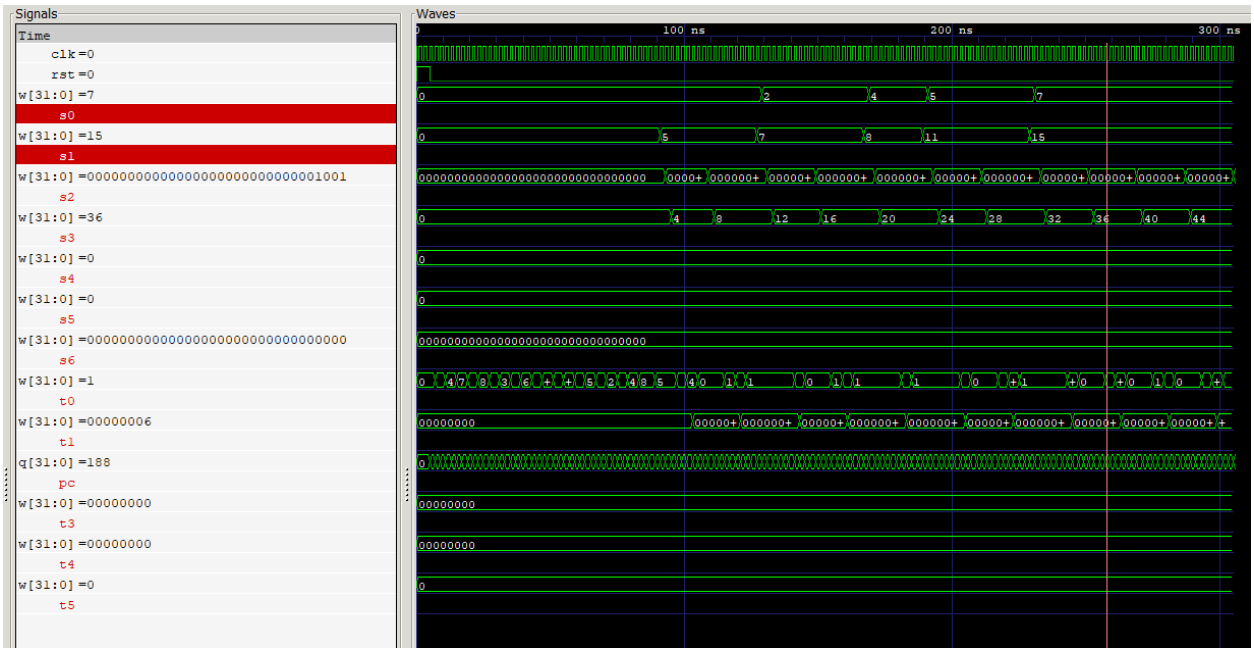
begin_for:
  slti t0, s2, 20
  beq t0, zero, end_for
  add t0, sp, s3
  lw t1, 0(t0)
  slt t0, s1, t1
  beq t0, zero, end_if
  add s1, zero, t1
  add s0, zero, s2
end_if:
  addi s2, s2, 1
  addi s3, s3, 4
  jal ra, begin_for
end_for:
```

کد زبان ماشین:

```
00000000000000000000000100010011
00000000010100000000001010010011
00000000010100010010000000100011
00000000010000000000001010010011
00000000010100010010001000100011
00000000011100000000001010010011
00000000010100010010010000100011
00000000001000000000001010010011
00000000010100010010011000100011
00000000010000000000001010010011
00000000010100010010100000100011
00000000010110000000001010010011
00000000010100010010101000100011
00000000001100000000001010010011
00000000010100010010110000100011
00000000011110000000001010010011
00000000010100010010111000100011
00000000001100000000001010010011
00000100101010001001000000100011
00000000100100000000001010010011
00000100101010001001000100100011
00000000110000000000001010010011
00000100101010001001001000100011
00000000111000000000001010010011
00000100101010001001001100100011
00000000101000000000001010010011
00000100101010001001010000100011
00000000110100000000001010010011
00000100101010001001010100100011
00000000001010000000001010010011
00000100101010001001011000100011
00000000000000000000001010010011
00000100101010001001011100100011
00000000001000000000001010010011
00000100010100010010000000100011
00000000001100000000001010010011
00000100010100010010001000100011
```

```
00000000010000000000001010010011
00000100010100010010010000100011
00000000010000000000001010010011
00000100010100010010011000100011
00000000000000000000001000110011
00000000000000001001000101000011
00000000000000010100001001011001 1
00000000000010000000010010001001 1
00000000010000000000100110010011
00000001010010010010001010010011
00000010000000101000010001100011
00000001001100010000001010110011
00000000000000101010001100000011
00000000011001001010001010110011
00000000000000101000011001100011
00000000011000000000010010110011
00000001001000000000010000110011
00000000000011001000010010001001 1
00000000010010011000100110010011
11111101100111111111000011101111
```

نتیجه شبیه سازی:



که اندیس ۷ و عدد ۱۵ را پیدا کرده است

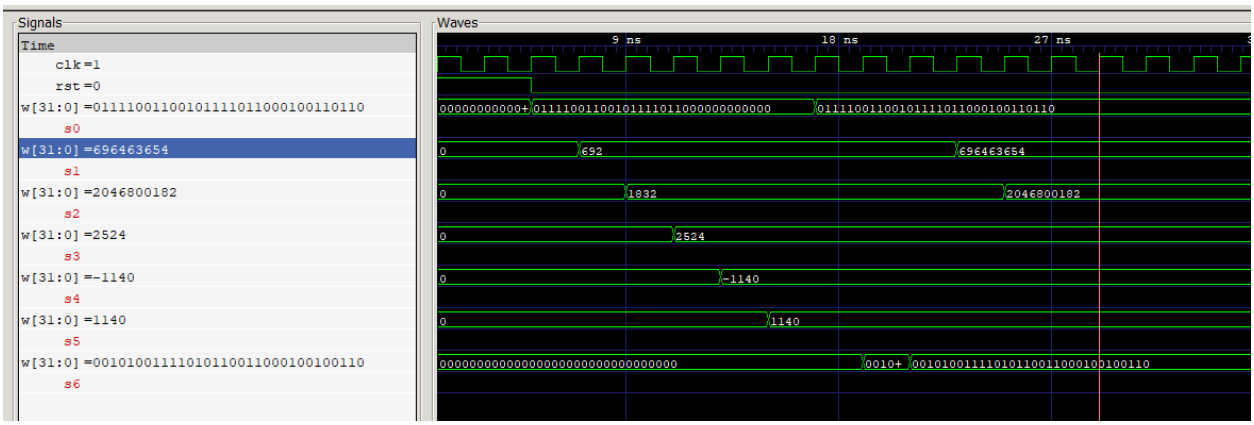
۲-۲- تست سایر دستورات

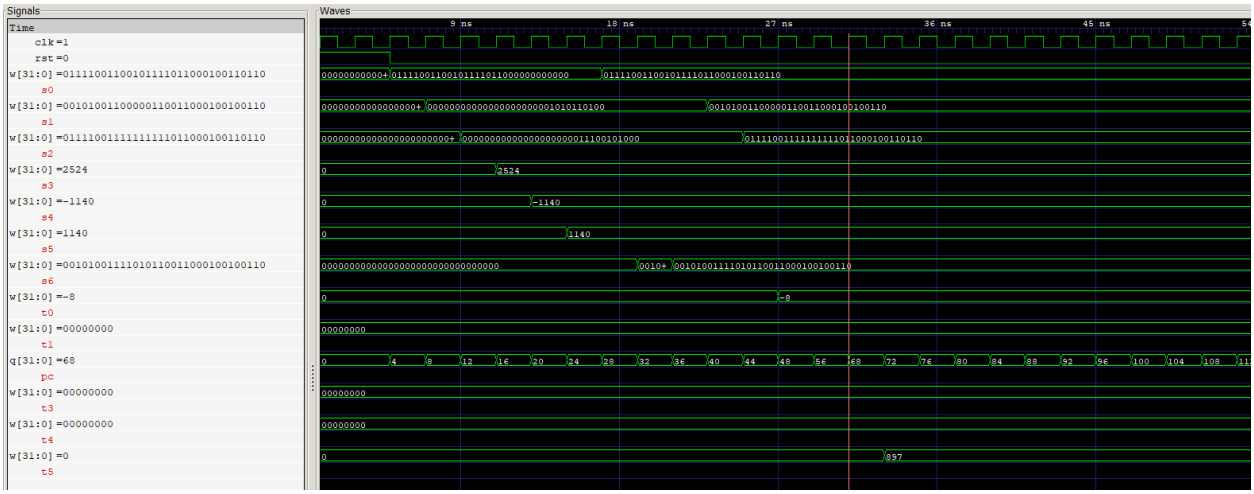
```
    lui s0, 498043 // 498043 = 01111001100101111011
    addi s1, zero, 692
    addi s2, zero, 1832
    add s3, s1, s2
    sub s4, s1, s2
    sub s5, s2, s1
    ori s0, s0, 310 // 310 = 000100110110
    lui s6, 171699 // 171699 = 00101001111010110011
    ori s6, s6, 294 // 294 = 000100100110
    and s1, s0, s6
    or s2, s6, s0
    addi t0, zero, -8
    jalr ra, t0, 64
    addi t1, zero, 88
    bne t0, t1, jj
    addi t3, zero, 932
    addi t4, zero, 1913
jj:
  addi t5, zero, 897
```

کد زبان ماشین:

```
01111001100101111011010000110111
00101011010000000000010010010011
01110010100000000000100100010011
00000001001001001000100110110011
01000001001001001000101000110011
01000000100110010000101010110011
00010011011001000110010000010011
00101001111010110011101100110111
00010010011010110110101100010011
00000001011001000111010010110011
00000000100010110110100100110011
11111111100000000000001010010011
00000100000000101000000011100111
00000101100000000000000001100010011
00000000011000101001011001100011
00111010010000000000111000010011
01110111100100000000111010010011
00111000000100000000111100010011
```

نتیجه شبیه سازی:





که صحیح هستند